

Adding non-timber values to harvesting planning: can the forestry sector do better for conservation?

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Abstract

Boreal landbirds, especially forest-associated ones, have been decreasing significantly since the early 70's. Using an aspatial LP-based harvest schedule optimization modelling framework, we tested the inclusion of a linear anticipation function that links forest landscape state to Red-breasted Nuthatch (*Sitta canadensis*) population density in Northern British Columbia. We demonstrate that this approach can be used to estimate the location and shape of Pareto-efficient tradeoff frontiers between harvest levels and bird population density. The method could potentially be applied to other locations, as well as other non-timber forest values.

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1 Introduction

As our planet currently faces severe climate, health and biodiversity crises (Ripple et al., 2020), identifying opportunities to act while maintaining economic growth and development becomes an intrinsic component of business across the globe (Addison et al., 2020). In Canada, forestry plays an important role in the country’s economy (Pappas et al., 2022) and might present itself as a favorable sector to explore methods to reduce the biodiversity impacts of current development. To integrate biodiversity conservation objectives into forest management and timber production, the forestry sector has been repeatedly proposing alternative silviculture methods in an attempt to partially offset its responsibility in the current biodiversity crisis. These strategies, however, are focused mostly on tree retention approaches (Ezquerro, Pardos, and Diaz-Balteiro, 2019). We propose here that working more directly with non-timber values might present the forestry sector with a unique opportunity to improve consequences of harvesting to wildlife.

Traditionally, conservation planning is done sequentially to harvesting planning, either before or after this. While an improvement for conservation objectives, such a strategy inevitably incurs a biodiversity opportunities’ loss, as Pareto dominated solutions are inefficient to reveal the essential trade-offs between the objectives. With this approach, directly assessing the tradeoffs between harvesting and biodiversity maintenance would be unattainable. We propose that, ideally, conservation planning should be placed inside of a timber supply model. Including non-timber values as objective functions within the harvest planning model could provide us with an improved understanding of tradeoffs between harvesting and biodiversity maintenance and new opportunities to reduce this sector’s footprint on the current crisis.

Boreal landbirds, especially forest-associated ones, have been decreasing significantly since the early 70’s (Rosenberg et al., 2019). Even though the causal link between these declines and forestry at the continent scale (Micheletti et al., in prep), there are undeniable consequences to bird species, especially mature-forest associated ones, in harvested lands. To demonstrate how our methods could improve the compromise between harvesting gains and biodiversity maintenance, we tested the inclusion of an anticipation function that defines forest association of the Red-breasted Nuthatch (RBNU; *Sitta canadensis*) in Northern British Columbia into the harvesting planning. Identified as moderately vulnerable (Ghalambor et al., 1999; Boreal Avian Modelling Project, 2020), RBNU nests mainly in mature conifer forests, habitat highly targeted by the forestry sector in Western Canada. We developed a set of scenarios, ranging from maximizing harvest revenue to maximizing RBNU abundance retention, and evaluate whether opportunities exist for the forestry sector to do better for a non-timber value while maintaining harvest production. The method presented here can be broadened and applied not only to other locations, but also other non-timber forest values, such as threatened species and carbon storage.

2 Methods

2.1 Study area and Species

The study area for our computational experiment is located in British Columbia (BC), Canada.

BC is the second leading province in the forestry sector, with about 77 million cubic metres harvested annually, of which 90% are under the Allowable Annual Cut scheme (Canada, 2020). Paradoxically, BC both leads biodiversity in Canada, harboring about 70% of mammal species and native Canadian breeding birds, and the most species assessed to be at risk (Westwood et al., 2019). One widespread bird species in Canada that has been deemed as moderately vulnerable to climate changes is the red-breasted nuthatch (RBNU; *Sitta canadensis*). With high forest association in BC, this small songbird inhabits dense conifer forests while avoiding broadleaf stands, making it an ideal species for the present study.

While the landbird model was fit using data from across the whole province due to data scarcity in our study area, the harvesting simulation was run on approximately 4.6 million hectares area, covering the Timber Supply Area (TSA 40) of Fort St John, in the Northeastern side of the Province.

2.2 Landbird model fitting

Open data on RBNU was requested to the eBird repository (Sullivan et al., 2009) and the received zipped file was hosted in private cloud repository. After this manual step, all following steps of the landbird model creation were developed in R (Team, 2019) were automated following the PERFICT approach (E. J. B. McIntire et al., 2022). The code that generated the bird model used for the harvesting simulation is openly hosted in GitHub (<https://github.com/tatimicheletti/fitBirdBiomassModel/tree/RIA>).

Firstly, we used the aboveground tree biomass and stand age layers from the Beaudoin et al. (2014) datasets as our initial tree biomass and tree stand age (hereby referred as KNN). For where data is available (32% of bird point counts), the KNN stand age layer was corrected with Forest Resources Inventory data, which is generated with on the ground measurements and presents, therefore, higher accuracy. The KNN layers were then cropped to the study area (BC province). The same procedure was done for a land cover layer (Latifovic and Pouliot, 2005). We then extracted the bird data from the eBird file (point count observations), aligned the bird data with the three vegetation layers, and extracted the biomass per species and land cover type for each bird count point. We then classified all individual tree species into softwood or hardwood and summed its biomasses. This generated a table with columns being observation counts, total biomass of softwood, total biomass of hardwood and stand age.

A generalized linear model was then fitted to the data as:

$$N_{bird} = B_{soft} + B_{hard} + A_{stand}, \quad (1)$$

where $Nbird$ is the observation counts of RBNU, $Bsoft$ and $Bhard$ is above-ground biomass of softwood and hardwood trees, respectively, and $Astand$ is forest stand age. This model generated the coefficients to be included in the anticipation function for the harvesting model, described below.

2.3 Harvesting model

The modelling framework used to simulate harvesting activity in this experiment was implemented using the **ws3** open-source Python package (Paradis, 2021c). **ws3** is a flexible software library that can be used to model complex landscape-level forest resource analysis scenarios, using datasets and model structures that are compatible with workflows used by many professional foresters in Canada to determine annual allowable cut levels for management units (MUs) located on public forested land. Integration of **ws3** into the **SpaDES** software environment (Chubaty and E. J. McIntire, 2019) is accomplished through the **spades_ws3_dataInit** and **spades_ws3** wrapper modules (Paradis, 2021b; Paradis, 2021a). Simulation result presented in this paper can be reproduced using code and data we published to a GitHub repository (Paradis, 2022a).

Our framework uses a *Model I* linear programming optimization problem formulation Johnson and Scheurman, 1977 to maximize even-flow harvest volume over an 80-year planning horizon, for each of 5 MUs.

The optimization problem can be formulated as follows

$$\max \sum_{i \in I} \sum_{j \in J_i} c_{ij} x_{ij} \quad (2)$$

s.t.

$$(1 - \varepsilon_p) y_p \leq \sum_{i \in I} \sum_{j \in J_i} \mu_{ijpt} x_{ij} \leq (1 + \varepsilon_p) y_p, \quad \forall p \in O', t \in T'_p \quad (3)$$

$$v_{ot}^- \leq \sum_{i \in Z} \sum_{j \in J_i} \mu_{ijot} x_{ij} \leq v_{ot}^+, \quad \forall o \in O, t \in T \quad (4)$$

$$\sum_{j \in J_i} x_{ij} = 1, \forall i \in Z \quad (5)$$

$$0 \leq x_{ij} \leq 1, \quad \forall i \in Z, j \in J_i \quad (6)$$

where

I := set of spatial zones
 J_i := set of available prescriptions for zone $i \in I$
 O := set of forest outputs
 $O' \subseteq O$:= set of targeted forest outputs
 T := set of time periods in the planning horizon
 $T'_p \subseteq O$:= subset of T on which even-flow constraints for output $p \in O'$ are applied
 ε_p := admissible level of variation on yield of targeted output $p \in O'$
 μ_{ijot} := quantity of output $o \in O$ produced in period $t \in T$ by prescription $j \in J_i$ in zone $i \in I$
 μ_{ijpt} := quantity of output $p \in O'$ produced in period $t \in T$ by prescription $j \in J_i$ in zone $i \in I$
 v_{ot}^- := lower bound on yield of output $o \in O$ in period $t \in T$
 v_{ot}^+ := upper bound on yield of output $o \in O$ in period $t \in T$
 c_{ij} := objective function contribution of prescription $j \in J_i$ in zone $i \in I$
 x_{ij} := proportion of zone $i \in Z$ on which prescription $j \in J_i$ is applied
 $y_p = \sum_{i \in I} \sum_{j \in J_i} \mu_{ijpt_p^R} x_{ij}$, i.e. total yield of targeted output $p \in O'$ at reference period t_p^R

The objective function (2) maximizes the sum of $c_{ij}x_{ij}$ products, which represent yield of a user-defined output—the output can be anything, but in the computational experiment presented here will represent typically harvest volume or bird abundance (depending on the context). The variables x_{ij} are linear, with domain $\{x_{ij} \in \mathbb{R} | 0 \leq x_{ij} \leq 1\}$ specified in (6). The set $O' \subseteq O$ represents targeted outputs, for which we enforce even-flow constraints. Even-flow constraints are defined in (3), and are expressed in terms of y_p , which represents total yield of targeted output $p \in O'$ in reference period t_p^R . Note that even-flow constraints are defined over time periods T'_p , which is any subset of T (need not be contiguous)— T'_p can be unique for each output $p \in O'$. General constraints (4) set upper and lower bounds on periodic yield of any output $o \in O$ —we use these constraints to set minimum TSA-wise levels of bird abundance to induce the scenarios presented in §3. Coverage constraints (5) require prescriptions to cover the entire zone—doing nothing for the entire planning horizon is considered a prescription that could generate some outputs.

We can use this model formulation to emulate a real-world optimized aspatial timber supply planning process, albeit with fewer constraints than the production-grade optimization models used in the timber supply review (TSR) process for the timber supply areas (TSAs) that constitute our analysis area.

Two key sources of input data for the forest estate models such as this are *initial inventory* and *yield curves*. We developed a reproducible data processing pipeline that automates the process of stratifying stands into analysis units

(AUs), and compiling yield curves for each AU. The process uses freely available data, freely available software, and some new open-source software components. The pipeline is implemented as a series of Jupyter Notebooks, and published online in a publicly available GitHub repository (Paradis, 2022b).

A general overview of the input data preprocessing pipeline follows. First, we import vegetation resource inventory (VRI) datasets and discard any stands outside of the study area, clean and filter the raw VRI data, and enhance it with additional information (e.g., site productivity, timber supply area membership, timber harvesting land base, etc.). Next, we aggregate stands with similar biogeoclimatic zone and leading species into strata, and then further subdivide each stratum into three AUs based on 30-point site index quantile ranges (low, medium and high), excluding data in the lowest and highest 5 quantile points for each stratum. For each AU, we generate an unmanaged (i.e., first growth) yield curve using VDYP (British Columbia Ministry of Forests, Lands, Natural Resource Operations and Rural Development, 2021). We model a single treatment option (clearcut, operable from age 40 onwards), which induces transition to pre-harvest yield curves at age 0. Note that we compiled a single monolithic model for the entire study area, which is composed of five timber supply areas (TSAs). The model includes TSA-wise even-flow constraints and TSA-wise bird abundance constraints, so it behaves like five independent timber supply models running in parallel.

We compiled one bird abundance yield curve $f_B(x)$ for each timber volume yield curve $f_T(x)$ using

$$f_G(x) = c_1 f_T(x) + c_2 f_T(x) + c_3 x + c_4$$

where x represents stand age $c_1 = +2.181 \times 10^{-3}$, $c_2 = -1.176 \times 10^{-2}$, $c_3 = -8.235 \times 10^{-4}$, and $c_4 = +7.594 \times 10^{-1}$ (coefficients derived from the regression analysis described in §2.2).

We used this modelling framework to generate a tradeoff curve for each of five TSAs, approximating the location and shape of the Pareto-efficient front describing the landscape-level relationship between harvest volume and bird abundance. We use an ϵ -constraint method (Haimes, 1971) to define the 11 scenarios that define the points along our tradeoff curves. Harvest volume is used as the reference objective function, and bird abundance is used for the ϵ -constraints. The 11 scenarios are induced by setting ϵ -constraint RHS values to evenly-spaced increments between nadir and maximum bird abundance indicator performance. The nadir bird abundance level is determined by maximizing harvest volume in the objective function, subject to even-flow harvest volume constraints. The maximum bird abundance level is determined by maximizing bird abundance in the objective function, subject to even-flow harvest volume constraints. We set $p = 0.10$ (i.e., allowed $\pm 10\%$ slack in the even-flow harvest volume constraint), and assumed 0.85 utilization rate when compiling harvest volume (i.e., only 85% of standing inventory, as defined by the merchantable volume yield curves, is counted as harvested volume).

The resulting LP primal problem matrix for each ϵ -constraint scenario has 6857 rows and 57771 columns (1653615 non-zero coefficient values). We solved

the optimization problems using Gurobi version 9.5.1rc2 on a linux server running Ubuntu 20.04.4 LTS, configured with 16 Intel Xeon Skylake processor cores (clocked at 2.49 Gz) and 180 Gb of RAM. Each optimization problem solves in approximately 2.44 seconds (not counting pre-processing and post-processing time, which takes several minutes per scenario).

2.4 Tradeoff analysis

The tradeoff curve between harvest volume and bird abundance was created by running eleven scenarios across the spectrum between maintaining as much bird abundance as possible, to maintaining as much harvesting as possible.

3 Results

Figure 1 shows simulation results, by TSA (subfigure rows). The leftmost column of subfigures shows harvest volume by scenario. The middle subfigure column shows bird abundance by scenario. The rightmost subfigure column shows harvest-bird tradeoff curves. x and y coordinates for tradeoff curve points represent the mean of scenario-wise data vectors for harvest volume and bird abundance, respectively.

Although our model predicts bird abundance on the entire landscape, the bird abundance data presented here was compiled only for the THLB portion of the landscape for each TSA so that both harvest volume and bird abundance data are based on the same area basis.

As explained in §2.3, we used the bird abundance indicator to define the ϵ constraints in our experiment. We defined the RHS values for the ϵ constraints such that the 11 scenarios in our computational experiment partition the bird abundance objective range (i.e., between nadir and maximum values) into 10 evenly-sized intervals, yielding evenly-spaced scenario-wise curves in the bird abundance (middle subfigure column) subfigures and an even step size along the x axis of the harvest-bird tradeoff (right subfigure column) subfigures.

The *cbird0.0* scenario is equivalent to the bird abundance nadir solution (i.e., bird abundance ϵ constraints absent or non-binding). At the other end of the spectrum the *cbird1.0* scenario sets bird abundance ϵ constraint RHS value to maximum feasible levels.

Setting up the ϵ constraints this way allows us to easily observe the systematically increasing marginal opportunity cost of increasing the bird abundance constraint RHS, where opportunity cost is measured in terms of reduction of harvest volume relative to the unconstrained AAC scenario. We see a similar pattern across all five TSAs of exponentially increasing opportunity cost as we progress through the sequence of ϵ -constrained scenarios—this can be seen in the steadily increasing gap between harvest volume scenario output (left subfigure column), as well as the convex shape of the tradeoff curves (right subfigure column).

A corollary of this is that all five TSAs exhibit a similar potential for substantial increases in bird abundance (relative to nadir levels) at relatively low marginal cost in terms of lost harvest volume—this can be shown by the relative proximity of the topmost (darkest) scenarios in the harvest volume subfigures (left column), as well as the relatively shallow (flat) slope at the far right of the tradeoff curves (right column). This pattern is most evident in TSAs 08, 40, and 41. We note that TSAs 16 and 24, which have a less convex (flatter) tradeoff response pattern, also have a much wider harvest volume opportunity cost range—harvesting essentially halts altogether in these TSAs when the bird abundance function is maximized.

In contrast with harvest levels, relative difference in bird abundance between scenarios *cbird1.0* and *cbird0.0* is quite low and stable across TSAs—in the 15–20% range—although the different patterns of bird abundance over time across TSAs suggests distinct landscape-level dynamics are at play in each subset of the study area.

4 Discussion

Discuss results here.

5 Conclusion

Insert concluding remarks here.

6 Acknowledgements

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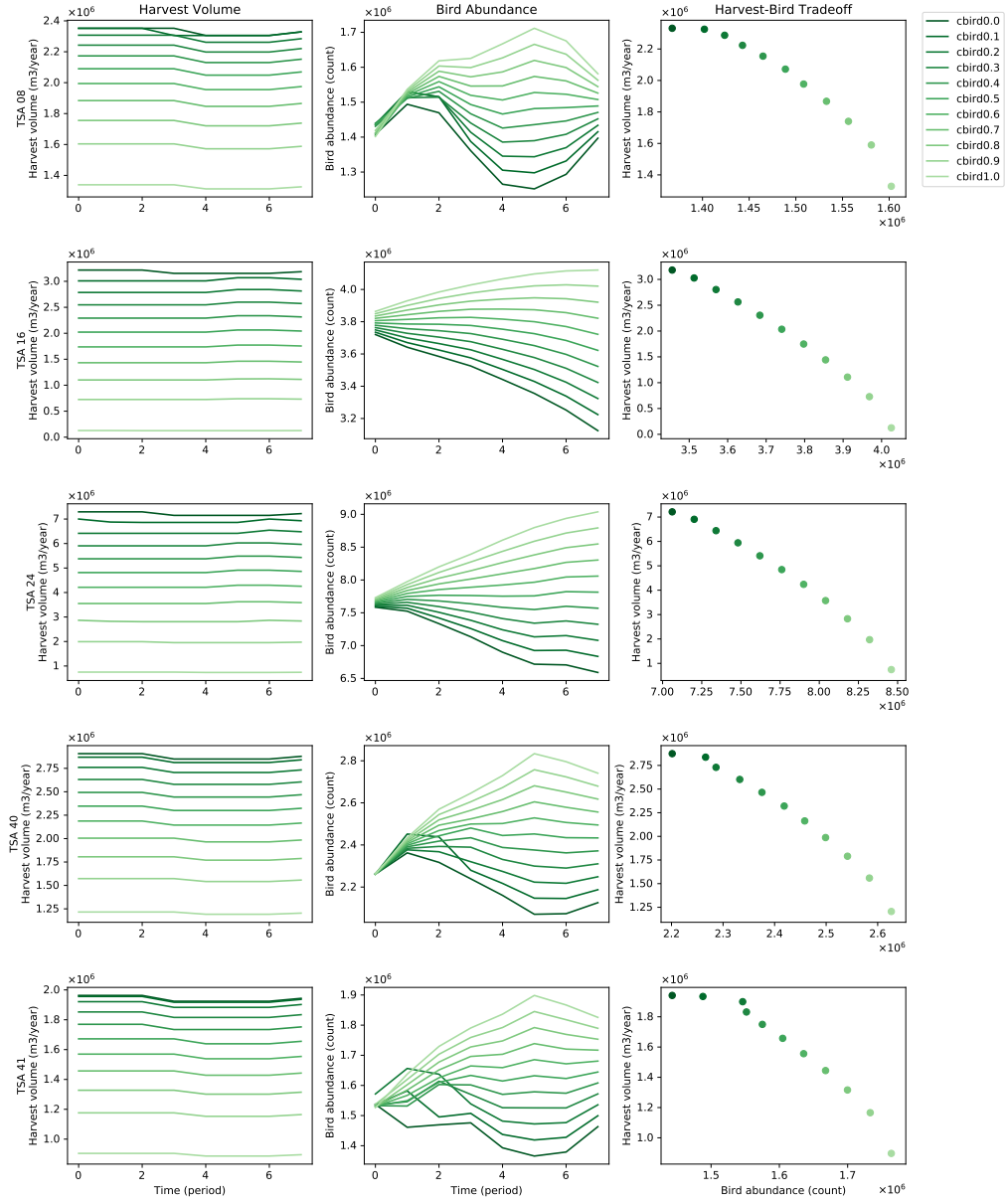


Figure 1: simulation results, by TSA (harvest volume, bird abundance, harvest-bird tradeoff).

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